

Artificial Intelligence, COMP 424, Fall 2026

Prerequisites:	COMP 206 or ECSE 321 MATH 323 or equivalent COMP 251	
Course schedule:	Mondays and Wednesdays, 4:05pm – 5:25pm SADB M-1 (08-31, 09-02) MCMED 522 (09-09 onwards)	
Instructors:	Jackie Chi Kit Cheung, jackie.cheung@mcgill.ca Office hours: Wednesdays, 9:30am – 11am in McConnell 108N Su Lin Blodgett, sulin.blodgett@mcgill.ca Office hours: Mondays, 2:00pm - 3:30pm in McConnell 312	
TA(s):	Shira Abramovich Aur�lie Gallet Sumukh Patel Mohamad Hosein Danesh Fahad Rahman Amik Kaixin Huang Audrey Le Meur (List is subject to change)	shira.abramovich@mail.mcgill.ca aurelie.gallet@mail.mcgill.ca sumukh.patel@mail.mcgill.ca mohamad.danesh@mail.mcgill.ca fahad.amik@mail.mcgill.ca kaixin.huang@mail.mcgill.ca audrey.lemeur@mail.mcgill.ca

Course Overview

Introduction to search methods. Knowledge representation using logic and probability. Planning and decision making under uncertainty. Introduction to machine learning.

Course Materials

Artificial Intelligence: A Modern Approach. Russell and Norvig. 4th edition. (Recommended)

You may purchase a copy of this textbook from the McGill bookstore or through an online retailer.

Learning outcomes

By the end of the course, students should have a broad understanding of the field of artificial intelligence. They should be able to define key terms and concepts in the field and be able to understand and implement some of the algorithms and techniques that underlie these technologies. They should have an awareness of the historical trends in artificial intelligence, as well as current issues and open problems.

Course Content

This list is tentative and subject to modifications.

- Search
- Game playing
- Logical reasoning
- Classical planning
- Probabilistic reasoning
- Learning probabilistic models
- Causal probabilistic models
- Utility theory
- Decision theory
- Sequential reasoning
- Applications
- Ethics and responsible artificial intelligence

Means of Assessment

Title	Weight	Description	Due Date	Considerations and Late Penalties
Midterm Exam	15%	In person, multiple choice and short answer.		Make-up exam to be held roughly one week later for students with a valid reason for missing the assessment.
Assignments	4 x 5%	Assignments to practice the concepts that are taught in the course.	Periodically throughout the term; roughly once per month	Late assignments accepted up to 24 hours after posted deadline without penalty.
Final Project	25%	Research project to be completed in a group of three.	Last day of class	Late submissions accepted up to 24 hours after posted deadline without penalty.
Final Exam	40%	In person, multiple choice and short answer during the final exam period.		According to McGill guidelines.

Assessments in this course are governed by the [Policy on Assessment of Student Learning](#) (PASL), which provides a set of common principles to guide the assessment of students' learning. Also see [Faculty of Science-specific rules](#) on the implementation of PASL.

Legally mandated academic accommodations are handled by Student Accessibility and Achievement. For more information see <https://www.mcgill.ca/access-achieve/>

In accord with McGill University's [Charter of Students' Rights](#), students in this course have the right to submit in English or in French written work that is to be graded. This does not apply to courses in which acquiring proficiency in a language is one of the objectives." (Approved by Senate on 21 January 2009)

Conformément à la [Charte des droits de l'étudiant](#) de l'Université McGill, chaque étudiant a le droit de soumettre en français ou en anglais tout travail écrit devant être noté, sauf dans le cas des cours dont l'un des objets est la maîtrise d'une langue. (Énoncé approuvé par le Sénat le 21 janvier 2009)

McGill University values academic integrity. Therefore, all students must understand the meaning and consequences of cheating, plagiarism and other academic offences under the [Code of Student Conduct and Disciplinary Procedures](#)" (Approved by Senate on 29 January 2003) (See [McGill's guide to academic honesty](#) for more information).

In the event of extraordinary circumstances beyond the University's control, the content and/or assessment tasks in this course are subject to change and students will be advised of the change.

Work submitted for this course must represent your own efforts. Assignments must be done individually; you must not work in groups unless otherwise stated. You must not copy any other person's work in any manner (electronically or otherwise), even if this work is in the public domain or you have permission from its author to use it and/or modify it in your own work. The only exceptions are for source code supplied by the instructor explicitly for an assignment, and for the final project, where usual research citation practices apply. Furthermore, you must not give a copy of your work to any other person.

Use of generative AI technologies. Unless otherwise specified by the instructor, you may use generative AI technologies (e.g., Chat-GPT, Gemini) only in an assistive manner (e.g., to help understand course content, to search for information, to help brainstorm ideas, or to check for grammar and style errors). You may not use such technologies as the primary means with which to complete your assignments or your final project. All substantive uses of these tools must be acknowledged in your submissions.

We may use automated software similarity detection tools to compare your assignment submissions to that of all other students registered in the course, and these tools are very effective at what they have been designed for. However, note that the main use of these tools is to determine which submissions should be manually checked for similarity by an instructor or TA; we will not accuse anyone of copying or working in groups based solely on the output of these tools. You may also be asked to present and explain your assignment submissions to an instructor at any time.

Cheating Vs. Collaborating Guidelines. Adapted from [Mausam's](#) and [Dan Weld's](#) guidelines.

Collaboration is a very good thing. On the other hand, cheating is considered a very serious offense. Concern about cheating creates an unpleasant environment for everyone. So how do you draw the line between collaboration and cheating? Here's a reasonable set of ground rules. Failure to understand and follow these rules will constitute cheating, and will be dealt with as per institute guidelines.

- **The 'The Office' Rule:** This rule says that you are free to meet with fellow students(s) and discuss assignments with them. Writing on a board or shared piece of paper is acceptable during the meeting; however, you should not take any written (electronic or otherwise) record away from the meeting. This applies when the assignment is supposed to be an individual effort or whenever two teams discuss common problems they are each encountering (inter-group collaboration). After the meeting, engage in a half hour of mind-numbing activity (like watching an episode of The Office), before starting to work on the assignment. This will assure that you are able to reconstruct what you learned from the meeting, by yourself, using your own brain.

- The Right to Information Rule: To assure that all collaboration is on the level, you must always write the name(s) of your collaborators on your assignment. This also applies when two groups collaborate.